DEPTH, DENSITY, AND OBJECTIVE ATP MEASUREMENT IN A FROZEN OCEAN

The L*=4000 benchmark as a calibrated proof-search environment

Working research note — Brian Tenneson — August 10, 2026

ABSTRACT

A proof-search problem can be deep without being sparse, sparse without being deep, or both deep and sparse. The frozen Ocean benchmark makes those distinctions measurable because its minimum proof depth is known externally and its entire directed search graph is finite. In the current L*=4000 family, no proof can reach the target in 3,999 or fewer benchmark inference steps, while a 4,000-step proof does exist. That establishes depth. The same instances also permit several different notions of density to be measured exactly: global graph density, the fraction of the graph that lies on any successful route, the fraction that lies on an optimal route, the density of the target in the first reachable shell, and the amount of branching ambiguity encountered along the unique shortest route.

This note argues that there should not be one undifferentiated number called “density.” It proposes an objective problem profile for a frozen proof ocean and a separate objective ATP profile for the navigator. It also gives a special role to Depths-F, the deliberately unfair or “trick” ATP. Depths-F should not be treated as a general-purpose competitor. It should be treated as a calibration floor: what happens when the navigation problem has effectively been removed and the solver merely has to traverse and emit the required proof. That control gives meaning to quantities such as navigation overhead, productive-inference precision, and logical penetration per primitive transaction.

1. THE FROZEN OBJECT OF STUDY

The primary object is not a processor, a Python program, or a stopwatch result. It is a frozen directed proof-search environment together with a frozen primitive notion of legal inference.

For the present Ocean family, let G=(V,E) be the directed graph encoded by the TPTP implications. There is one source s, one target t, and one benchmark inference for every directed edge. A proof is a directed path from s to t. The benchmark generator plants four successful routes of lengths

4000, 4003, 4007, and 4011,

then adds dead-end burden. Node labels are randomly permuted. The benchmark is checked independently by breadth-first search before the ATP sees it. For all 20 completed L*=4000 instances, the externally verified minimum distance is

$$L^* = d(s,t) = 4000.$$

The benchmark artifact from the completed professional-ATP run contains, for every seed,

$|V| = 88,019$ vertices,
$|E| = 88,021$ directed edges.

The current six-way workflow seals the same family by stripping the L* comment, replacing informative filenames, and reordering/renaming the visible axioms. Those operations change presentation order but preserve the directed graph, so the problem-side quantities defined below are unchanged.

Notation caution. Earlier Ocean notes use F(n) for transparent chains and F(n,w,...) for hidden-route oceans. In this note, “the L*=4000 Ocean” means the hidden-route frozen family whose minimum source-to-target path is 4000. If F(4000) is retained for the transparent chain, the hidden-route family should continue to be written F(4000,w,...) or O(4000) to avoid conflating the calibration chain with the search ocean.

2. DEPTH: WHAT 4000 MEANS

Depth is the cleanest quantity in this benchmark.

Define

$L^* = \min\{|P| : P \text{ is a valid directed proof path from } s \text{ to } t\}$.

For every one of the 20 frozen instances,

$L^* = 4000$.

Therefore:

- the target is absent from every ball $B_r(s)$ for $r \leq 3999$;
- no ATP, regardless of intelligence or hardware, can produce a valid benchmark proof with fewer than 4000 primitive edge-inferences under this frozen presentation;
- a 4000-edge certificate is an objective lower bound on proof cost;
- any returned proof of length L has an objective route-quality ratio $Q=L/L^*$, with $Q=1$ exactly for an optimal proof.

The statement “cannot be reached in 3999 steps” is therefore not a timeout statement and not an empirical guess. It is a graph-theoretic fact about the frozen benchmark.

This makes L* a problem quantity, not an ATP quantity.

3. DENSITY IS NOT ONE NUMBER

The word density can refer to several genuinely different objects. They should be reported separately.

3.1 Global directed graph density

The ordinary directed graph density is

$$\delta_G = |E| / (|V|(|V|-1)).$$

For the L*=4000 Ocean,

$$\delta_G = 88{,}021 / (88{,}019 \times 88{,}018) \approx 1.13616 \times 10^{-5} \approx 0.001136\%.$$

By this conventional graph-theoretic measure the ocean is extremely sparse: only about eleven directed edges exist per million possible ordered vertex pairs.

This is a property of the presentation graph. It is not yet a measure of how much of the graph is useful for reaching the target.

3.2 Solution-support density

Define the solution-support set

$$V_{sol} = \{v \in V : t \text{ is reachable from } v\}.$$

A vertex belongs to V_{sol} exactly when the search has not yet fallen irreversibly into a dead-end region. Reverse reachability from t computes this set exactly.

Across every L*=4000 seed,

$$|V_{sol}| = 16{,}019,$$

so

$$\rho_{sol,V} = 16{,}019 / 88{,}019 \approx 18.1995\%.$$

Likewise 16,021 of the 88,021 edges lie within the successful-route structure:

$$\rho_{sol,E} = 16{,}021 / 88{,}021 \approx 18.2013\%.$$

The complement is especially simple:

72,000 vertices and 72,000 edges are dead-end burden.

Thus approximately 81.8% of the graph, by either vertex or edge count, does not lie on any route that can eventually reach the target.

This is much closer to the intuitive Ocean question: how much of the water contains routes that can still reach land?

3.3 Optimal-route density

The shortest route is unique in these generated instances. Its support contains

4,001 vertices and 4,000 edges.

Therefore the global fraction of the graph lying on the unique shortest proof is

$\rho_{opt,V} = 4{,}001 / 88{,}019 \approx 4.5456\%$,

$\rho_{opt,E} = 4{,}000 / 88{,}021 \approx 4.5444\%$.

This is not the probability that an arbitrary ATP will find the optimum. It is a structural baseline: roughly one edge in twenty-two belongs to the unique minimum-length route.

3.4 Target density at the first reachable horizon

Let

$S_r(s) = \{v : d(s,v)=r\}$

be the exact depth-r shell, and

$B_r(s) = \{v : d(s,v)<=r\}$

be the cumulative ball.

At r=3999, the target density is exactly zero because t is not in B_{3999}.

At r=4000, the target appears for the first time. Across the 20 seeds, the size of the first reachable shell S_{4000} ranges from 15 to 34 states, with median 22. Therefore the target's density in the first reachable shell has median

$1/22 \approx 4.545\%$,

with a range from about 2.94% to 6.67% across seeds.

There are exactly four solution-support states in S_4000: the target itself on the shortest route plus one state on each of the three longer successful routes. Thus the median solution-support density of the first reachable shell is

$$4/22 \approx 18.18\%.$$

The cumulative ball is much larger. B_4000 contains between 87,836 and 87,927 vertices, with median 87,877.5. The target is one state in that entire reachable region, so the median cumulative target density at the first possible proof horizon is approximately

$$\begin{aligned} 1 / 87{,}877.5 &\approx 1.13795 \times 10^{-5} \\ &\approx 0.001138\%. \end{aligned}$$

That gives a useful contrast:

• depth says the target cannot exist before radius 4000;
• shell density says that once radius 4000 is reached, the target is one of roughly 22 frontier states;
• cumulative target density says that among everything reachable by then, only one state is the designated target.

These are all objective, but they answer different questions.

4. BRANCHING AMBIGUITY ALONG THE SHORTEST PROOF

Depth alone does not say how difficult it is to stay on the right route. A 4000-step chain with one outgoing edge at every state is deep but navigationally trivial. A 4000-step route with repeated branching can be much harder to follow.

For the unique shortest path P*, define the branch-information quantity

I_branch(P*) = sum over v in P* before t of log2(outdeg(v)).

This is entirely determined by the frozen graph. Its probability interpretation, however, requires an additional assumption: a navigator that chooses uniformly among visible outgoing edges at each step.

Across the 20 L*=4000 oceans:

• the median shortest route has 3,387.5 branching points among its 4,000 transitions, about 84.69% of the route;
• the median mean outdegree along the shortest route is about 2.7845;

• I_branch has median approximately 5,293 bits, ranging from about 5,225 to 5,367 bits.

Under the deliberately naive local-uniform interpretation, an uninterrupted walk that always chooses the unique shortest continuation would have a median probability on the order of

10^-1593.

That number must not be advertised as a universal “probability of proving the theorem.” Sophisticated ATPs do not choose uniformly, and exhaustive methods do not rely on one uninterrupted lucky walk. The useful point is narrower: the frozen graph contains thousands of bits of local branching ambiguity along the optimal route even though the final target itself is only one state among about 88,000 reachable states.

This exposes why a single density number is inadequate. The cumulative target scarcity is only about log2(87,878) ≈ 16.42 bits, while the sequential local-choice burden along the shortest path is about 5,293 bits under the local-uniform model. One measures a static target set; the other measures a sequence of branching decisions.

5. THE “TRICK ATP” AS A CALIBRATION FLOOR

The simple specialized ATP, called Depths-F in the earlier Ocean work, should be retained precisely because it is unfair.

Depths-F is not meant to demonstrate general theorem-proving intelligence. It understands the special successor/reachability structure well enough that the navigation problem is effectively removed. The earlier Technical Companion therefore describes it as a calibration ceiling or known-map control rather than a fair competitor.

That makes it scientifically valuable.

If an idealized Depths-F returns the unique optimal proof in this benchmark, several objective quantities are fixed immediately:

Minimum proof depth:
$L^* = 4000$.

Returned proof length:
$L = 4000$.

Route quality:
$Q = L/L^* = 1$.

Certificate cost in primitive Ocean inferences:

$C = 4000.$

No valid solver can beat $C=4000$ while the frozen benchmark continues to define one graph edge as one primitive proof inference.

The difficult question is what to call its search work. If the specialized program already has the map encoded in its method, it may perform only a few high-level software decisions before emitting the path. It would be misleading to compare those high-level decisions with Vampire clauses, E given clauses, or Data-ATP inference attempts as though all were the same unit.

A better canonical accounting is therefore:

$W = C + A_{off},$

where C is the number of primitive inference edges in the final verified certificate and A_{off} is the number of additional primitive candidate inference applications explored but not used in that certificate.

For the ideal trick ATP,

$C = 4000,$
$A_{off} = 0,$
$W = 4000.$

Its logical navigation-overhead factor is then

$\Omega = (W-L^*) / L^* = 0.$

Its productive-work fraction is

$P = C/W = 1.$

And a simple logical penetration efficiency can be written

$\eta = L^*/W = 1.$

These are not claims that the program uses zero CPU time or performs only 4000 machine instructions. They are claims about the frozen logical accounting system. Physical implementation has been factored out.

This is exactly why the trick ATP belongs in the benchmark as a control. It defines a visible floor:

4000 unavoidable proof inferences, zero unnecessary logical detours.

A general-purpose ATP can match the floor only if it finds the optimal route without spending counted primitive work elsewhere. It cannot legitimately go below the 4000-inference certificate requirement.

6. DENSITY ENRICHMENT FOR THE CALIBRATION ATP

The trick ATP also gives a clean interpretation of “density enrichment.”

Globally, only about 18.2013% of edges belong to any successful route. If every counted inference made by Depths-F is productive, its productive-edge fraction is 1. Relative to the global successful-edge baseline, that is an enrichment factor of approximately

$$G_{sol} = 1 / 0.182013 \approx 5.49.$$

Relative to the unique optimal-route edge density,

$$G_{opt} = 1 / 0.0454437 \approx 22.01.$$

These ratios should be labeled as global-reference enrichment factors, not universal ATP scores. A sophisticated ATP sees local information, not a uniformly sampled global edge. Still, they make the calibration role concrete: the specialized control concentrates essentially all of its logical work into a region occupying only about 4.54% of the graph’s edges if optimality is required.

7. OBJECTIVE PROBLEM PROFILE VERSUS OBJECTIVE ATP PROFILE

The cleanest framework is to keep problem quantities and solver quantities separate.

Problem-side Ocean profile

O(G,s,t) = (
  L*,
  |V|,
  |E|,
  δ_G,
  ρ_sol,
  ρ_opt,
  |S_L*|,
  |B_L*|,
  I_branch,
  dead-end fraction,
  ...
).

For the present L*=4000 suite, a representative profile is:

L* = 4000;
|V| = 88,019;
|E| = 88,021;
global directed density ≈ 0.001136%;
solution-support vertex density ≈ 18.1995%;
solution-support edge density ≈ 18.2013%;
optimal-route edge density ≈ 4.5444%;
first-reachable shell size median = 22;
first-horizon cumulative ball size median = 87,877.5;
median branch information ≈ 5,293 bits;
dead-end burden ≈ 81.8%.

ATP-side logical profile

For a solver policy π on the frozen ocean, report at least:

success;
L and Q=L/L*;
C, the primitive cost of the final independently verified certificate;
A_off, counted primitive search work not appearing in the final certificate;
W=C+A_off;
Ω=(W-L*)/L*, logical overhead above the unavoidable optimum;
P=C/W, productive-work fraction;
best proof horizon reached as a function of W;
logical penetration or horizon reduction per unit W;
minimum-cost certification work, if separately performed.

Physical profile

Only after those quantities are reported should the implementation layer be added:

wall-clock time;
RAM;
processor and runner;
parallelism;
energy or other engineering quantities.

The physical profile matters, but it answers a different question.

8. WHY COMPLICATED ATPs ARE HARDER TO COMPARE

Vampire, E, SPASS, Prover9, Data-ATP, and DATA 2.0.1 do not naturally expose the same internal primitive counter.

A saturation prover may process a clause and generate many inferences. A portfolio prover may score thousands of possibilities before choosing one. A learned ATP may spend significant computation ranking a small number of logical expansions. A proof-horizon certifier may deliberately repeat exhaustive work in order to establish optimality. A specialized control may encode the answer structure in its policy and spend almost no navigation work at all.

Therefore raw native counters must remain native counters unless there is a defensible normalization.

The common ground is the frozen certificate language. Every successful solver can be required to return a certificate that reduces to the same primitive Ocean edge-inferences. That makes L, L*, Q, and certificate cost C genuinely comparable.

Search work is harder. The strongest next step is to define a canonical logical transaction model and instrument each solver adapter to map its native actions into that model. Until that is done, a table should not silently equate:

one Vampire inference record,
one E given-clause cycle,
one Data-ATP expansion,
one Depths-F high-level decision.

They are different objects.

9. WHAT DEPTHS-F SHOULD DO IN THE NEXT BENCHMARK

Depths-F probably should have appeared beside the professional ATPs and the two DATA reference implementations, but nothing is lost by adding it in the next frozen benchmark version.

It should be labeled explicitly:

DEPTHS-F — SPECIALIZED CALIBRATION CONTROL — NOT A GENERAL-PURPOSE ATP.

It should receive the same sealed problem representation unless the experiment intentionally defines a separate known-map calibration condition. Its special knowledge must be declared. The result should not be merged into a single “winner” ranking.

Its role is to answer a different question:

How close is a general ATP to the logical floor once the unavoidable 4000-step certificate cost is separated from navigation overhead?

That gives every other ATP a meaningful reference point.

If Depths-F has $W=4000$ and Vampire, for example, ultimately requires a larger canonical logical-work count W_V on the same frozen instance, then the ratio

$$W_V / 4000$$

is an interpretable logical overhead factor once both counts have actually been normalized to the same primitive transaction system. The stopwatch is no longer needed to define that comparison.

10. RESEARCH QUESTIONS SUGGESTED BY THE $L^*=4000$ OCEAN

The present benchmark suggests several questions that are more informative than “which ATP was fastest?”

First, does L^* predict search difficulty once solution-support density and branching information are held fixed or varied independently?

Second, can two oceans have the same $L^*=4000$ but radically different ρ_{sol}, ρ_{opt}, shell width, and I_{branch}, producing radically different ATP rankings?

Third, does a learned ATP increase productive-work fraction P or density enrichment G while reducing logical overhead Ω?

Fourth, can a horizon curve $H^*(W)$ distinguish an ATP that makes steady logical progress from one that spends almost its entire budget far from any proof and then suddenly lands?

Fifth, how strongly do physical rankings differ from logical-work rankings? A solver may be slow per expansion but make excellent logical decisions, while another may perform huge numbers of cheap expansions per second.

Sixth, what happens if the decoy factor is changed while $L^*=4000$ is held constant? The current family makes this especially attractive because depth and dead-end burden can be manipulated separately.

Seventh, how should the same quantities be generalized from this deliberately simple implication graph to natural theorem libraries, where the minimum proof length may be unknown and the proof-state graph cannot be enumerated completely?

11. CENTRAL CONCLUSION

The L*=4000 Ocean already gives more than a stopwatch benchmark.

It gives an environment in which depth, density, width, dead-end burden, shortest-route support, and branching ambiguity can all be separated and measured. The target is literally impossible before 4000 primitive proof steps. About 81.8% of the graph is dead-end burden. Only about 4.54% of edges lie on the unique shortest proof. At the first reachable depth, the target is one state among a median frontier of 22, while the cumulative reachable region already contains roughly 87,878 states. Along the shortest route, thousands of branching decisions create a very different notion of scarcity than static target density alone captures.

The specialized Depths-F control then supplies the logical floor. It cannot make the 4000-step proof shorter, but it can make navigation overhead essentially disappear. Under a canonical logical accounting, that means C=4000, W=4000, Ω=0, P=1, and η=1. Those numbers are useful precisely because the ATP is a trick.

The deeper benchmark question is therefore not merely:

Did the ATP find the proof?

It is:

How much of the frozen logical ocean did the ATP have to spend before it reached a proof whose unavoidable depth was already fixed by the problem itself?

Depth tells us how far land is. Density tells us how much useful territory surrounds the route. Branch information tells us how ambiguous the local navigation is. Canonical logical work tells us how efficiently the navigator crossed the ocean. Physical time can then be reported separately as the engineering cost of realizing that abstract search.

DATA SOURCE AND REPRODUCIBILITY NOTE

The numerical L*=4000 measurements in this note were computed from all 20 TPTP instances preserved in GitHub Actions run 31412655224, artifact “ocean-professional-atp-results-L4000,” generated from btenneson/ATP commit 3f7d4a345dbada286e5b16b46b787408189ecbef. For each instance, the analysis reconstructed the directed graph, ran BFS from the source, ran reverse reachability from the target, identified the unique shortest-path support using d(s,v)+d(v,t)=4000, measured exact shells and balls, and summed log2(outdegree) along the shortest route. The benchmark generator independently verifies L*=4000 before producing each instance.

The current six-way benchmark uses a sealed, presentation-reordered version of the same graph family. Results from that workflow should be added later as solver-side measurements; they are not assumed in the problem-side calculations above.

RELATED INTERNAL SOURCES

Brian Tenneson, The Proof Horizon and the Ocean of Proof — Technical Companion, August 2026.

Brian Tenneson, DATA 2.0.1: From a Target Formula to a Verified Minimum-Cost Proof, Version 2.0.1 Research Draft, August 9, 2026.

Data-ATP repository, HILBERT_THEOREM_SEARCH_IMPLEMENTATION.md, including the frozen definition of one Data-ATP expansion as one attempted rule application on one valid decoded premise tuple.

btenneson/ATP, benchmarks/ocean/generate_ocean_tptp.py, frozen Ocean generator.